Homework 4

AMATH 582/482, Winter 2026

Due on Mar 2, 2026 at midnight.

Directions, Reminders and Policies

Read these instructions carefully:

- **You are required to upload a PDF report to Gradescope along with your code in the form of a Jupyter notebook or a zip file of your main scripts and functions. The report and the code are to be submitted under separate assignments. DO NOT INCLUDE YOUR CODE IN THE REPORT.**
- The report should be a maximum of 10 pages in accordance with the template on Canvas including references. Minimum font size 11pts and margins of at least 1inch on A4 or standard letter size paper. The report should be formatted as follows:
  - Title/author/abstract: Title, author/address lines, and short (100 words or less) abstract. This is not meant to be a separate title page.
  - Sec. 1. Introduction and Overview
  - Sec. 2. Theoretical Background
  - Sec. 3. Algorithm Implementation and Development
  - Sec. 4. Computational Results
  - Sec. 5. Summary and Conclusions
  - Acknowledgments. Mention other students if you were part of a study group.
  - References
- **Recall our late policy: If you plan to use your tokens please send an email to Saba (`heravi@uw.edu`). If you do not use a token you will lose 1/3 of your grade for each day the submission is late. WE WILL NOT AUTOMATICALLY APPLY TOKENS. YOU MUST INFORM US IF YOU PLAN TO USE THEM.**

Problem Description: Classifying Red Wine

Your goal in this homework is to build a classifier for red wine quality. The data set provided for this homework is a subset of a larger data set that is available on the UCI Machine Learning Repository at `https://archive.ics.uci.edu/ml/datasets/wine+quality`. The data set contains 1599 samples of red wine, each with 11 features, including various chemical properties, and a quality score between 0 and 10; the file `wine_description.txt` contains a detailed description of these features.

You will consider this problem as a multi-class classification problem with 11 classes corresponding to the quality scores and build various classifiers using affine models and kernel ridge regression with multiple kernels and compare the test performance of your models. You are required to use the training and test splits provided in the files `wine_training.csv` and `wine_test.csv`. You also need to implement appropriate hyper-parameter tuning strategies to best fit your models to the data.

Tasks

Below is a list of tasks to complete in this homework and discuss in your report.

1. Train an affine classifier using the one-hot encoding method and report the test performance. Explain clearly how you implemented the one-hot encoding method, what optimization problem you solved, and how you fitted the affine model.
2. Implement kernel ridge regression classifiers using the one-hot encoding method with the RBF, Laplace, and polynomial kernels:

$$k_{\mathrm{RBF}}(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right), \quad k_{\mathrm{Laplace}}(x, x') = \exp\left(-\frac{\|x - x'\|}{\sigma}\right), \quad k_{\mathrm{Poly}}(x, x') = (x^T x' + c)^d,$$

   and report the test performance. Explain clearly how you implemented your kernel regression model and how you chose the hyper-parameters. For the polynomial kernel you may try different values of c and d and report the best performance you obtained.
3. Discuss how the performance of the various models compare to each other and what you think are the reasons for the differences in their performance.
4. Use your three models to predict the quality of the wines in the file `wine_new_batch.csv` and report the predicted quality scores in a table.

Hint: Don't forget to standardize and center the features before fitting your models as the kernel regressors can be very sensitive to the scale of the features otherwise.